

A novel framework for optimization and evaluation of sensors network in urban drainage system

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ABSTRACT

Efficient management of urban drainage system (UDS) is crucial for understanding the operating status of UDS and facilitating urban flood early warning. Establishing an appropriate sensor network is fundamental to achieving cost-effective and sustainable management of UDS. Previous researches have predominantly focused on optimizing sensor placement but have often overlooked the evaluation of sensor network performance. To address this gap, we propose a framework that not only optimizes sensor placement using information theory but also evaluates the performance of sensor networks through matrix completion. After the method was tested in the case study, we found that the information amount provided by the selected nodes and information redundancy among nodes in UDS can both be effectively represented by the information theory approach, and then optimal sensor networks with different numbers of sensors was selected. Furthermore, the matrix completion algorithm successfully evaluated the sensor network's performance in operation status perception and flooding risk assessment. The results indicated that the operation status perception error was 33 %, and the flooding risk assessment accuracy reached 76 % with four sensors. Increasing the sensor count to eight reduced the error to 29 % and improved accuracy to 82 %. Thus, it is evident that the matrix completion algorithm is a rapid and accurate method for evaluating sensor network performance. This study provides a comprehensive framework for sensor network optimization and evaluation, which can greatly facilitate the development of urban flood risk early warning and sustainable management of UDS.

1. Introduction

The rapid urbanization and climate change have aggravated the frequency of extreme rainfall events and the impact of urban floods (Mabrouk et al., 2024; Wu et al., 2023). The Urban Drainage System (UDS) is a critical infrastructure that transports the collected stormwater to receiving water bodies for mitigating flooding severity (Wang et al., 2023). In combined system, raw effluents are spilled directly into surface and causing serious pollution on the environment during extreme rainfall events (Diaz-Fierros et al., 2002; Even et al., 2004). Thus, understanding the status of urban drainage system is very important for flood warning and environmental management. Sensor network is the important basis for obtaining information characteristics of the UDS, whether it is used for continuous monitoring of flood-prone points (Li, 2021), model calibration (Huang et al., 2022) or monitoring the status of equipment (Miao et al., 2018). Theoretically, a comprehensive approach

is to monitor dynamic operating conditions and detect anomalies of UDS through installing sensors at all nodes. However, the number of sensors is generally limited by constraints such as budget, installation and maintenance (Meijer et al., 2018). It is very important to design an adequate sensor network to efficiently reduce the potential economic losses and public health damage caused by urban flooding and overflow pollution.

The problem of sensor optimization in UDS has been widely studied in recent years. The previous studies have mainly focused on urban flooding scenario and illicit intrusions of pollutants. For example, the soluble contaminants (such as heavy metals or soluble ionic compounds) were used as indicator parameters for the optimal placement of water quality sensors in urban drainage systems to identify illicit intrusion (Banik et al., 2017; Sambito et al., 2019; Yazdi, 2018). Fattoruso et al. (2015) optimized the design of sensor network for flood forecasting modelling with the goal of maximizing the calibration model accuracy.

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Moreover, the optimal sensor layout was found by using the conventional pollutant (chemical oxygen demand, COD) and hydraulic characteristics (water level and overflow volume) to meet the demand for routine monitoring of UDS (Wang et al., 2023). It is necessary to choose appropriate indicators or objective functions based on the application scenarios of designed sensor networks. More importantly, since various complex challenges with UDS during operation (sediment behavior, clogging or sensor mechanical damage by debris), which can easily cause damage to sensor equipment (Xu et al., 2019), suitable sensor monitoring indicators for selection are often limited. In contrast, the information obtained from water level time series can effectively provide the operating status and overflow risk of nodes in UDS during the rainfall events, and the water level sensor is stable and suitable for long-term monitoring, especially non-contact water level sensor.

Aside from the selection of monitoring indicator, the sensor layout optimization design method is essential to the construction of an efficient sensor network. Clustering analysis method has been used in the field of sensor placement optimization, such as water distribution system (Khaksar Fasaee et al., 2021), river system (Lee et al., 2020), and reservoir system (Mu et al., 2023). However, the clustering results depend heavily on algorithm selection and parameter setting, and its subjectivity may affect the final designed sensor network (Nardo et al., 2018). Conversely, the intensive application of information theory in monitoring network design and assessment of water resources systems, this approach can objectively and accurately evaluate the connections and redundancies among variables. The sensor optimization methods based on information theory have also been successfully applied to contamination early warning of water distribution system (Khorshidi et al., 2018), reservoir water quality monitoring (Pourshahabi et al., 2020), and contaminant detection of UDS (Yazdi, 2018), but this approach has been slightly used in UDS particularly for routine monitoring and urban flood risk.

Furthermore, there is still no rapid and accurate quantitative evaluation method for the designed sensor networks. Although previous research proposed an indirect quantification and comparison method based on joint entropy for monitoring network evaluation (Wang et al., 2023), this indicator still cannot accurately evaluate the performance of the designed sensor networks. Specifically, probability of detection and average time to detection were often used as indicators in water distribution system to assessing the performance of designed sensor network for contamination illicit intrusion or leakage (Khorshidi, 2020; Khorshidi et al., 2018). To verify the performance of the optimized sensor networks in reservoir for water quality monitoring, the mean values of water quality index of time series were compared (Pourshahabi et al., 2020). Thus, a similar method is urgently developed to evaluate the designed sensor network after the optimization layout of sensors.

Matrix completion algorithm is a technology used to process missing data. Its principle is to infer the value of missing data based on known data to obtain a complete matrix, and it is often applied to image inpainting, collaborative filtering and traffic data imputation (Fan and Cheng, 2018; Fan and Chow, 2017). The matrix completion had also been used for low-cost data collection for sensor networks in weather monitoring and water treatment (Savvaki et al., 2015; Xie et al., 2018). The main factor affecting the accuracy of the matrix completion model is the amount of information contained in the known elements of matrix (i. e., the uncovered parts of the image in image restoration problem) (Monti et al., 2017; Zhang et al., 2020). Therefore, if the selected monitoring points are regarded as known elements of original matrix, the performance of designed sensor networks can be effectively evaluated by analyzing the results of matrix completion model. To the best of the authors' knowledge, the capability of matrix completion to support sensor network evaluation has not been investigated by the previous researches, and there are no studies that use this approach for sensor network design (including urban drainage systems).

Real-time monitoring of the whole urban drainage system is an important means of assessing urban flooding risk. It is of great

significance to obtain the operation status perception of UDS by optimizing the layout of limited water level sensors. Therefore, this study proposes a novel framework for sensor network optimization and evaluation based on information theory and matrix completion. The main objectives of this research are as follows: (1) Optimize the sensor placement of UDS for flooding risk monitoring, (2) Evaluate and quantify the performance of designed sensor networks. We believe this study can significantly contribute to addressing urban flooding challenges, promoting the development of sustainable cities.

2. Methodology

The proposed framework consists of three steps as shown in Fig. 1. Firstly, a numerical theoretical model is developed to simulate operation process and pluvial flooding of UDS under different rainfall scenario and parameter combination. Then, the model simulation results are used to optimize sensor placement based on information theory and intelligent optimization algorithm. Finally, the matrix completion algorithm is applied in Step 3 using sensor placement optimization results to evaluate the effectiveness of sensor network.

2.1. Urban drainage system simulation

The urban drainage flood simulation is performed using the Storm Water Management Model (SWMM) by U.S. EPA (Rossman, 2010), which is one of the most commonly used models in rainfall-runoff response and flooding simulation of UDS (Mugume et al., 2015; Wu et al., 2023). A flooding event simulation is associated with a variety of uncertainties, the most significant of which originate from rainfall events and model parameters (Huang et al., 2022; Lin et al., 2022). For this purpose, return periods, rainfall duration and rainfall peak position coefficient are considered as the parameters to generate rainfall events. The rainfall intensity is calculated using the local storm intensity equation, and the Chicago approach is used to redistribute the rainfall amounts before and after the rainfall peak position (Chen et al., 2018). Subsequently, Monte Carlo Simulation (MCS) algorithm is applied to incorporate the uncertainties emanating from the SWMM model parameter combination. According to the previous researches, twelve common input parameters of the SWMM model are used as the normal distribution-based inputs in MCS (Huang et al., 2022; Liu et al., 2023; Swathi et al., 2019). It is worth mentioning that the expected value of the normal distribution is the parameter after the SWMM model calibration, while the standard deviation is determined based on the parameter sensitivity of the literature survey (Liu et al., 2020; Ma et al., 2022). The descriptions of selection parameters for MCS and the corresponding abbreviation are listed in Table S1. In summary, a generated rainfall events and b random model parameter combinations are selected to simulate $a \times b$ urban drainage flood scenarios (Fig. 1). The results of the simulated scenarios are then used to calculate the flooding at each node. This approach can provide a more comprehensive and objective understanding of the flooding risks of each node in UDS. These results are used to derive the information theory process and optimize the sensor placement for the UDS.

2.2. Sensor placement optimization

2.2.1. Information theory

Information theory is a powerful method for analyzing uncertainty in water resources systems when information is restricted. Shannon (1948) demonstrated that events with high frequency convey less information, but events with low probability yield more information. Uncertainty is reduced as new knowledge is obtained, and the value of new information is proportional to the degree of uncertainty reduction. Therefore, information theory can be used as an approach to quantify the uncertainty and knowledge about the flood characteristics of the UDS. In this study, we use the value of information (VOI) and transinformation

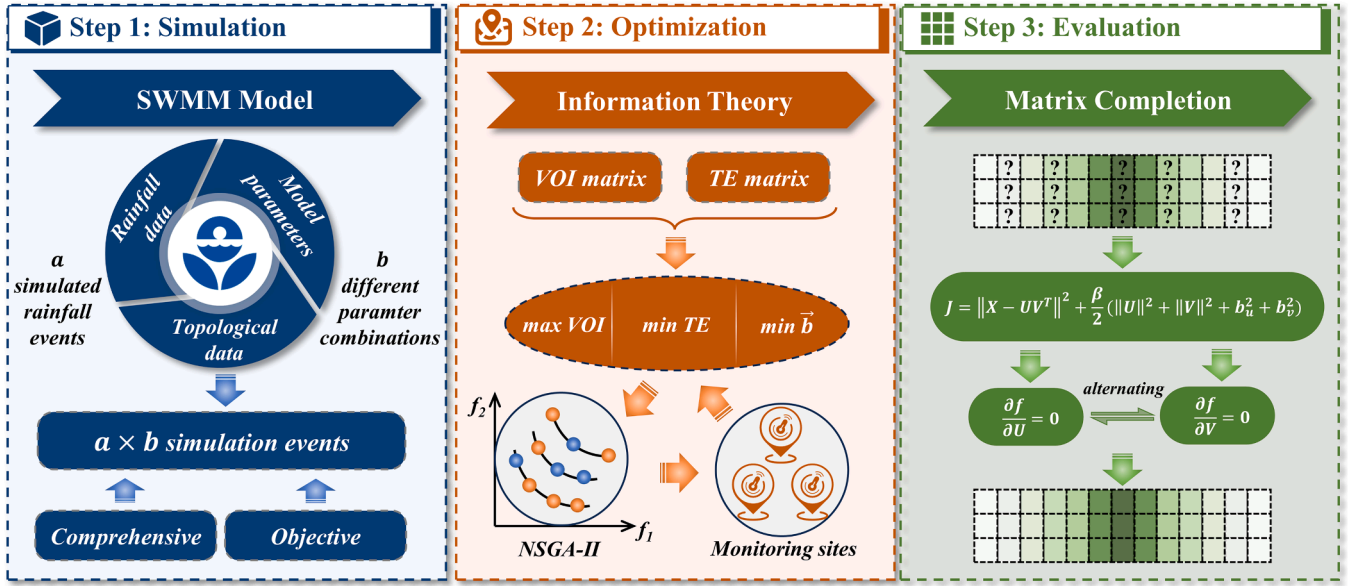


Fig. 1. Overview of the proposed sensor placement optimization and evaluation framework for urban flood.

entropy (TE) to assess the value and uniqueness of information at various nodes in UDS. The VOI can determine the amount of information in a set of nodes, while the TE can quantify the information redundancy among these nodes (Pourshahabi et al., 2020). The details are given below.

The state s of each node can be defined as s_1 (Normal) and s_2 (Flooding), which can be quantified in discrete form as the vector of prior probability $P(s)$. The information received from the sensors are q_1 (Relax) and q_2 (Danger). Based on this theory, the belief can be updated according to the new information from the UDS. The specific equation can be expressed as:

$$P(s|q) = \frac{P(q|s)P(s)}{P(q)} \quad (1)$$

Where, $P(s|q)$ is the updated belief after receiving the new information m ; $P(q|s)$ is the conditional probability of receiving the new information m when the state of the node is s ; and $P(q)$ is the probability of the receiving new information q , called evidence probability.

Once the prior probability $P(s)$ is determined, it is necessary to obtain new information independent of the prior belief to calculate the evidence probabilities $P(q)$, and the evidence conditional probabilities $P(q|s)$ before calculating the updated belief $P(s|q)$. For this purpose, we assume a sensor is placed at node i , and probabilities of flooding risk is calculated by the new information. Then, in order to obtain $P(q|s)$, pair the selected node i with another node j , and calculate the conditional probabilities of receiving flooding warning from the selected node i and the detection states of the other node j , respectively. In this study, the prior and evidence probabilities are calculated using the simulation results of random scenarios, which are independent of the Bayesian perspective due to their random generation (Alfonso et al., 2016; Alfonso and Price, 2012).

From the over $a \times b$ simulated urban drainage flood scenarios in section 2.1, about 80 % are used for calculating the prior probabilities, and the remaining 20 % are used for calculating the evidence probabilities. We presume the municipal managers use a sensor placed at node i to warn the citizens located at node j about the flooding risk. Timely warnings would preclude major disaster, while the false positive and false negative would result in panic or exposure of citizens, respectively (Khorshidi et al., 2018). Hence, the consequences, $C(a, s)$, of executing an action a given a particular state s of node j , form a matrix that contains the costs associated with choosing to do an action based on the state the

decision maker believes is occurring. Table 1 shows the the consequences matrix $C(a, s)$, where the negative values indicate a cost. The cost of issuing a flood warning is estimated to be around \$5 for communication and mobilization, and \$30 for false alarms. The cost of doing nothing while there is flooding is \$100 (a high cost due to the disaster associated with flooding that is not warned), and it is \$0 if there is no flooding (Alfonso and Price, 2012). The usefulness of those actions based on the received message, u_m , and without the received message, u_s , can be represented as:

$$u_m = \sum_s C(a, s)P(s|q) \quad (2)$$

$$u_s = \sum_s C(a, s)P(s) \quad (3)$$

where, S is the total number of flooding states, s . Supposing that the rational decision maker will always choose the action with the maximum utility, therefore, the VOI of node i for determining the flooding state of node j , the final expression for $VOI_i(j)$ can be calculated as:

$$VOI_i(j) = \sum_M P(q) \left[\max_a \left(\sum_s C(a, s)P(s|q) \right) - \max_a \left(\sum_s C(a, s)P(s) \right) \right] \quad (4)$$

where, M is the number of information. For the node i , VOI value can be calculated for all other nodes j , which will result in a VOI_i curve. The purpose of this study is to identify the sensor place i that can describe the states of the maximum number of node in UDS. A common approach is adding up the individual values of $VOI_i(j)$ of all the nodes in UDS to obtain the total VOI of node i for the state of the entire water system. Thus, it is corresponding to the area under the VOI_i curve when the infinite number of selected sensors in UDS. VOI_i can be estimated as:

Table 1

Arbitrary consequence matrix of doing action a given state s (Costs Units).

$C(a, s)$	a_1	a_2
s_1	-5	-100
s_2	-30	0

$$VOI_i = \int_j VOI_i(j) dl \quad (5)$$

where, l is the spatial dimension of the selected sensor places. The best place for a sensor is where it provides reliable information about the status of the maximum number of nodes in UDS. Therefore, it is recommended to place the sensor at the position where it has the largest VOI_i , namely,

$$\max\{VOI_i\} = \max\left\{\int_j VOI_i(j) dl\right\} \quad (6)$$

However, a significant number of sensors influence the level of information redundancy. Hence, the mutual information, also known as transinformation $TE(i, j)$, of two selected nodes i and j for sensor placement can be determined as follows (Watanabe, 1960):

$$TE(i, j) = \sum_{s_1} \sum_{s_2} P(i_{s_1}, j_{s_2}) \ln \left[\frac{P(i_{s_1}, j_{s_2})}{P(i_{s_1})P(j_{s_2})} \right] \quad (7)$$

where, $P(i_{s_1}, j_{s_2})$ is the joint probability of having the state s_1 at node i , while the state of node j is s_2 ; $P(i_{s_1})$ and $P(j_{s_2})$ are the probabilities of having the states s_1 and s_2 at node i and j , respectively. The mutual information indicates the amount of information that each of the chosen sensors shares, and accounting for the interdependencies among their partial combinations.

2.2.2. Multi-objective optimization

The optimal placement of the monitoring sensors have been done through evaluating three objectives as shown in Step two of Fig. 1. Firstly, maximize the total VOI value of the selected node in UDS to ensure that the most information is obtained. Secondly, the total TE value of the selected nodes in the UDS is minimized to ensure the minimum redundant information. Finally, minimize the number of monitoring sensors to reduce associated economic costs (assuming that the number of sensors has a positive correlation with the cost). Mathematically the optimization problem can be formulated as:

$$\text{maximize } Z_1 = \frac{1}{\max\{[VOI]\}} \sum_{ij} \max\{b_i \times VOI_i(j)\} \quad (8)$$

$$\text{minimize } Z_2 = \frac{1}{\max\{[TE]\}} \sum_{ij \neq i} \max\{b_i \times b_j \times TE(i, j)\} \quad (9)$$

$$\text{minimize } Z_3 = \sum_{vi} b_i \quad (10)$$

where, b_i, b_j are a vector of binary decision variables to find the optimal sensor placement. As above, this optimization problem is solved by considering all possible combinations of the N locations in UDS, which indicates that it is a typical NP-complete problem. Thus, the Non-dominated Sorting in Genetic Algorithms (NSGA-II) is used to solve this optimization problem, which is a powerful multi-objective intelligent optimization algorithm (Srinivas and Deb, 1994).

2.3. Sensor network evaluation

To further objectively evaluate the perception ability of the selected monitoring sensors combination on the operating status and overflow risk of the nodes in UDS, it is crucial to quantify the impact of the value of information and transinformation entropy. If the impact of sensor placement optimization on the perception capability of UDS is not quantified, or if the increase of VOI (decrease of TE) cannot effectively improve the perception capability, the rationality and effectiveness of

the sensor placement optimization cannot be proven. Therefore, a simple and effective test method is necessary to quantify the perception capability after sensor placement optimization, which also can be used to balance the number of sensors and economic costs.

We regard the nodes at the selected monitoring sensor places as known elements, and the remaining nodes as unknown elements to develop the water level missing matrix $X \in \mathbb{R}^{M \times N}$ (Fig. 2). The aim of this section is to find an approximate matrix of the missing matrix, fill in the missing elements of the matrix with the corresponding water level values in the approximate matrix, and thus obtain the water level reconstructed matrix $\tilde{X} \in \mathbb{R}^{M \times N}$. Hence, the amount of effective information contained in the selected monitoring sensors has a significant impact on the accuracy of the water level reconstructed matrix. Finally, we transform the problem of evaluating the perception capability of sensor network into a typical matrix completion problem in mathematics.

The premise for solving the matrix completion problem is that the original matrix has low rank. Because the low-rank matrix contains a lot of redundant information, which can restore the complete matrix through some elements for the original matrix (Candès and Recht, 2012; Koren et al., 2009). Non-Negative Matrix Factorization (NMF) is a common method to solve matrix completion problem, the purpose of the standard NMF is to find two non-negative matrices whose product is an optimal approximation to the original matrix (Sotiras et al., 2015; Xu et al., 2015). Therefore, we can obtain the following standard optimization problem as shown in Fig 2.

$$\begin{aligned} \text{minimize } J &= \|X - \tilde{X}\|^2 \\ &= \|X - UV^T\|^2 \\ &= \sum_{i,j, x_{ij} \neq \text{nan}} \left(x_{ij} - \sum_{l=1}^k u_{il} v_{jl} \right)^2 \end{aligned} \quad (11)$$

where, $U \in \mathbb{R}^{M \times k}$, $V \in \mathbb{R}^{N \times k}$ are randomly initialized non-negative matrix; i, j are the row and column of matrix X , respectively. It is worth noting that the loss function J is only calculated using the known elements in the missing matrix to avoid information leakage. There are some errors and noises in the actual observation data matrix of UDS. Hence, it is necessary to add regularization terms and bias terms to optimize the loss function.

$$\begin{aligned} \text{minimize } J &= \|X - \tilde{X}\|^2 + \frac{\alpha}{2} (\|U\|^2 + \|V\|^2) + \frac{\beta}{2} (b_u^2 + b_v^2) \\ &= \sum_{i,j, x_{ij} \neq \text{nan}} \left(x_{ij} - \sum_{l=1}^k u_{il} v_{jl} \right)^2 + \frac{\alpha}{2} \left(\sum_{i,l} u_{il}^2 + \sum_{j,l} v_{jl}^2 \right) + \frac{\beta}{2} \left(\sum_i b_{u,i}^2 + \sum_j b_{v,j}^2 \right) \end{aligned} \quad (12)$$

where, α, β are the regularization coefficient and bias coefficient, respectively; $\|U\|, \|V\|$ are the regularization terms; b_u, b_v are the bias terms. The NMF algorithm is a typical convex optimization problem that involves solving two matrix variables U and V , so the efficient alternating least squares algorithm is used to solve the problem in this study (He et al., 2018). The initialization of $\|U\|$ and $\|V\|$ use the standard normal distribution, and all the algorithms are programmed and calculated using the Python language. The exact solution derivation procedure and the associated pseudo-code are in the supplementary material.

Due to the initialization of $\|U\|$ and $\|V\|$ have a certain degree of randomness, we use the optimal selected monitoring sensors combination to calculate water level results of all nodes through NMF algorithm, and repeat the same process 100 times for each monitoring sensors combination to eliminate the influence random initialization of parameters. In addition, the effectiveness of the sensor placement optimization method is evaluated by randomly selecting the same number of monitoring sensors combination and performing the same water level

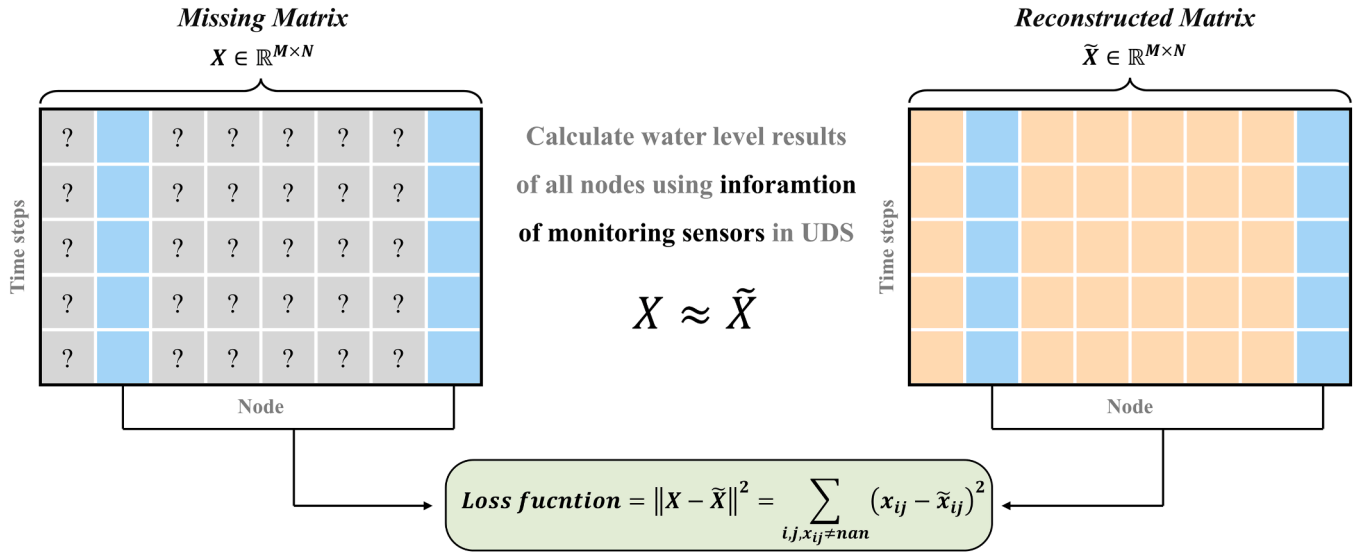


Fig. 2. The schematic diagram of matrix completion.

calculation process of nodes in UDS.

3. Case study

3.1. Study area

HZ city is located in the southeast coastal district of China, which has a subtropical monsoon climate. The eastern region is densely covered with river networks, and the annual rainfall is approximately 1454 mm. The study area is a residential area near the Qiantang River, and there is a small river crosses the whole study area (Fig. 3). The total area of the study area is 2.679 km², with 878 nodes and 898 conduits (after simplification). The study area is divided into 2693 subcatchments to simulate the urban drainage system under different rainfall scenarios. Furthermore, we use the rectangular open conduits to model the small river that crosses the study area. There is one rain gauge station in this area, and three existing sensors (S1-S3) are used to monitor the water level of the nodes in UDS (Fig. 3). The study region often suffers from

pipe network backflow and urban flooding due to extreme rainfall weather. It is of great significance to obtain the operation status of urban drainage system and flooding risk in time for early warning and reducing disaster losses. The performance of sensor networks is the basis to achieve this goal. Therefore, it is a suitable place to study the optimal sensor placement for urban flooding.

3.2. SWMM model calibration and validation

The sensor placement optimization approach based on information theory needs to understand the operating status and flooding risk of the UDS under different scenarios. Due to the influence of model input parameters, the simulation results of the SWMM model without calibration are generally completely inconsistent with the actual drainage system (Huang et al., 2022). Thus, we can preliminarily eliminate the impact of the parameter uncertainty through model calibration. We selected twelve main SWMM input parameters to be calibrated as mentioned in section 2.1, and the ranges of value were shown in Table S2. After

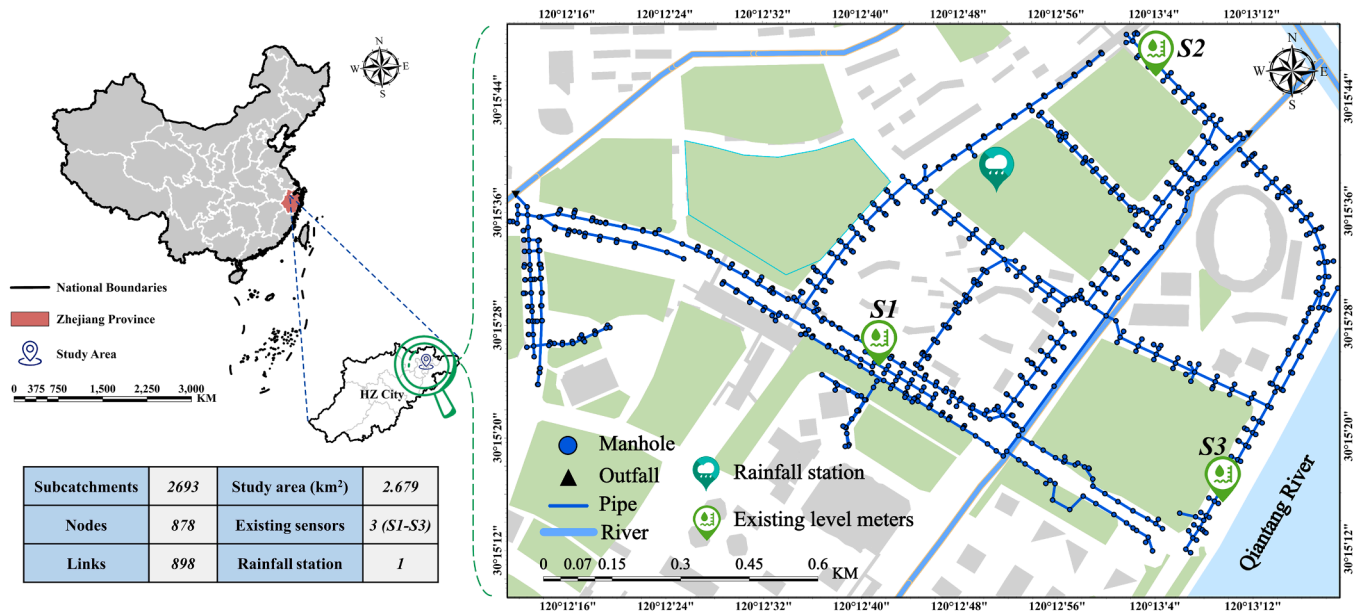


Fig. 3. The urban drainage system (UDS) of study area.

defining the parameters, a real rainfall event was chosen for the model calibration and validation in this study, as shown in Table S3. According to the records of rainfall station, the total rainfall of this rainfall event was 16.7mm, the maximum rainfall intensity was 19.8 mm/h, and the total duration was 480 min. The water level sensor S1 was used to calibrate the SWMM model to find the optimal input parameter combination, and the remaining sensors (S2 and S3) were selected for model validation. If the model cannot be calibrated and validated due to lack of observed data on water level or flow, it is recommended to verify the rationality of the SWMM simulation results through recorded flood markers or using the runoff coefficient method (Li et al., 2023).

It can be seen from the Fig. S1, the simulation results of three sensors are overall similar to the observed data during this rainfall event. The Nash-Sutcliffe efficiency coefficient (NSE) is used to evaluate the accuracy of the SWMM model in this study. According to the current assessment standard for hydraulic model, the NSE value of the model calibration and validation should be ≥ 0.5 . The NSE value between the simulation results and observation results in S1 was 0.907, and the values of S2 and S3 for validation were 0.948 and 0.901, respectively. This indicated that the developed model possesses acceptable accuracy, and it can simulate the hydrodynamic process of the study area. Hence, it can be considered that the influence of the uncertainty of SWMM model parameters on scenario simulation of UDS has been preliminarily eliminated.

3.3. Data collection and preprocessing

According to the first step of proposed framework, we used different combination of return periods, rainfall duration and rainfall peak position coefficient to generate rainfall events for model simulation. The local rainfall intensities were calculated as:

$$q = \frac{1455.550 \times (1 + 0.958 \lg P)}{(t + 5.861)^{0.674}} \quad (13)$$

where, q is the design rainfall intensity, $L/s \cdot hm^2$; P is the return period, a ; and t is the rainfall duration, min . Specifically, we selected six different return periods (1a, 2a, 5a, 10a, 20a, and 50a), eight different rainfall duration (30min, 60min, 120min, 180min, 240min, 360min, 720min, and 1440min), and three different rainfall peak position coefficient (0.2, 0.5, and 0.8) to generate 144 rainfall events. Then, 300 different combination of SWMM parameters were sampled using the MCS algorithm. All parameters were used a normal distribution, and their mean values (the model parameters after calibration) and standard deviations were shown in Table S2. Hence, a total of 43200 simulation were performed under different scenarios of the UDS based on the combination of rainfall events and input parameters.

In this study, two real rainfall events were provided for sensor network evaluation of UDS. Details of these rainfall events were presented in Table S3. The event occurred between 04:25 on 11 December 2023 and 08:00 on 11 December 2023 was a rainfall scenario without node flooding in UDS. Another rainfall event with a peak rainfall intensity of 89.7 mm/h caused overflow in a series of nodes, and resulting in a serious urban flooding. Therefore, these two rainfall events were used to evaluate the sensor network performance of operation status perception and flooding risk assessment, respectively.

3.4. Method application and evaluation

The proposed framework was applied in the study area for sensor placement optimization and evaluation. From the over 43200 simulated flooding scenarios in UDS, about 80 % were used for calculation of prior probabilities, and the remaining 20 % were used for calculation of evidence probabilities. The VOI matrix and TE matrix were calculated according to Eq. (4) and Eq. (7). The decision space could be significantly enhanced, due to the VOI and TE calculation summarizes all scenarios in this two matrices. Hence, more accurate optimization algorithm (NSGA-

II) was used to find the optimal combination of nodes through three objective functions (Eq. 8-10) in our framework. Moreover, VOI and TE approaches significantly reduce the memory requirements and provide faster runtimes, especially in an optimization process. It is necessary to calculate the rank of the original matrix to determine the feasibility of the matrix completion algorithm. We extracted the original matrix X of the 43200 simulations, and the singular value decomposition was used to calculate the rank of each matrix.

$$X = \sum_{i=1}^M s_i u_i v_i^T \quad (14)$$

where, the singular values s_i satisfy the relationship of $s_1 \geq s_2 \geq \dots \geq s_M$; u_i and v_i are the left singular vector and right singular vector, respectively. Then the E_k can be calculated by the following equation:

$$E_k = \frac{\sum_{i=1}^k s_i}{\sum_{i=1}^M s_i} \quad (15)$$

where E_k is the proportion of singular values of the first k order. As shown in Fig. S2, the mean of first two singular values accounted for 99.5 %. This result indicated that the original matrix X is low-rank matrix. Subsequently, we used matrix completion approach to evaluate the performance of sensor network under two typical scenarios. All the programs were developed in this study using the Python 3.9. The implemented programs were run on a personal computer with an AMD Ryzen 7 5700G and 64 GB random access memory, with a NVIDIA RTX 3060Ti 8 GB graphic processing unit (GPU).

Two metrics were selected here for evaluating the matrix completion performance of operation status perception under the no flooding scenario. These are the root mean square error (RMSE) and the mean absolute percentage error (MAPE), which have been widely used in the urban drainage system simulation. The calculation equations for the RMSE and MAPE are as follows:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (16)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{y_i} \right| \quad (17)$$

where, y_i is the observation data of water level; $\hat{y}_i$ is the calculation value of water level. Another three classification evaluation metrics were used to evaluate the matrix completion performance of flooding risk assessment under the flooding scenario. The accuracy score is one of the most widely used indices for classification and means the percentage of the number of correctly predicted and total samples (Wang et al., 2024). It can be computed with Eq. (18):

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (18)$$

where TP is the true positives, TN is the true negatives, FP is the false positives, and FN is the false negatives. To solve the problem of sample imbalance, Recall and False Positive Rate (FPR) were selected for evaluation as follows:

$$Recall = \frac{TP}{TP + FN} \quad (19)$$

$$FPR = \frac{FP}{FP + TN} \quad (20)$$

4. Results and discussion

4.1. Results of information theory calculation

From the perspective of information theory, the VOI matrix represents the value of information that each node can provide, while the TE matrix represents the degree of information redundancy. Specifically, a value of 1 in the i th row and j th column of the VOI matrix indicates that regardless of whether the sensor is placed at node i or node j , it will provide the same value to the municipal managers. On the other hand, a value of 0 would indicate that placing a sensor at node i would not provide any information about the flooding risk of node j . The same element in the TE matrix would indicate complete information redundancy, if it is equal to 1, and complete information uniqueness if equal to 0 (Khorshidi, 2020). To provide a graphical example of this interpretation, three nodes (denoted as 1, 2, and 3) with different flooding risk probabilities (scenario probability of flooding for all simulated scenarios at each node) of 0, 0.50 and 0.87 were selected, and the normalized VOI (TE) and the relationship between VOI (TE) and node flooding probabilities were shown in Fig. 4 and Fig. S3-S4. It is vivid in sub-figures a and b of these figures that the VOI and TE were not reliant on spatial distance, and they showed a regular distribution of the VOI values. This finding was not consistent with the previous studies that optimized water quality sensor placement for early warning in case of illicit water intrusion in sewer systems or water distribution systems (Banik et al., 2017; Khorshidi, 2020; Khorshidi et al., 2018). Because the intrusion

location of contaminant was stochastic variable, the smaller the distance from the water quality sensor, the greater the amount of information that can be provided, which is also accompanied by the increase of information redundancy. But there are some nodes with stable state under various rainfall events in UDS, some of which are nodes with almost no flooding, while the others have a high risk of flooding (called flood-prone points). This characteristic led to the fact that the VOI distribution was relatively regular at different sensor places, and had no spatial dependence in this study. Moreover, the distribution of TE values were different under the case of the three selected typical nodes, and the overall value increased with flooding risk probability, but it is still did not depend on the spatial distance. These results indicated that the characteristics of rational sensor networks were different according to the different monitoring tasks. Hence, the specific sensor placement optimization strategy should be considered depend on the required monitoring purpose and perception object in practice.

To further explore the values of VOI and TE of nodes with different flooding risk probabilities, Fig. 4c and 4d depicted the relationship between flood probability and VOI (TE) of node 1. As shown in the figure, as the flood probability increased, the normalized VOI increased linearly and then decreased linearly, and the normalized VOI was the largest when the flood probability was about 0.5. There were the same results in nodes 2 and 3 according to Fig. S3-S4. These results indicated that the information provided by sensor node i was the most beneficial for the node j with flooding probability of 0.5. A reasonable explanation for this result is that it is due to the state of the node with a flooding probability

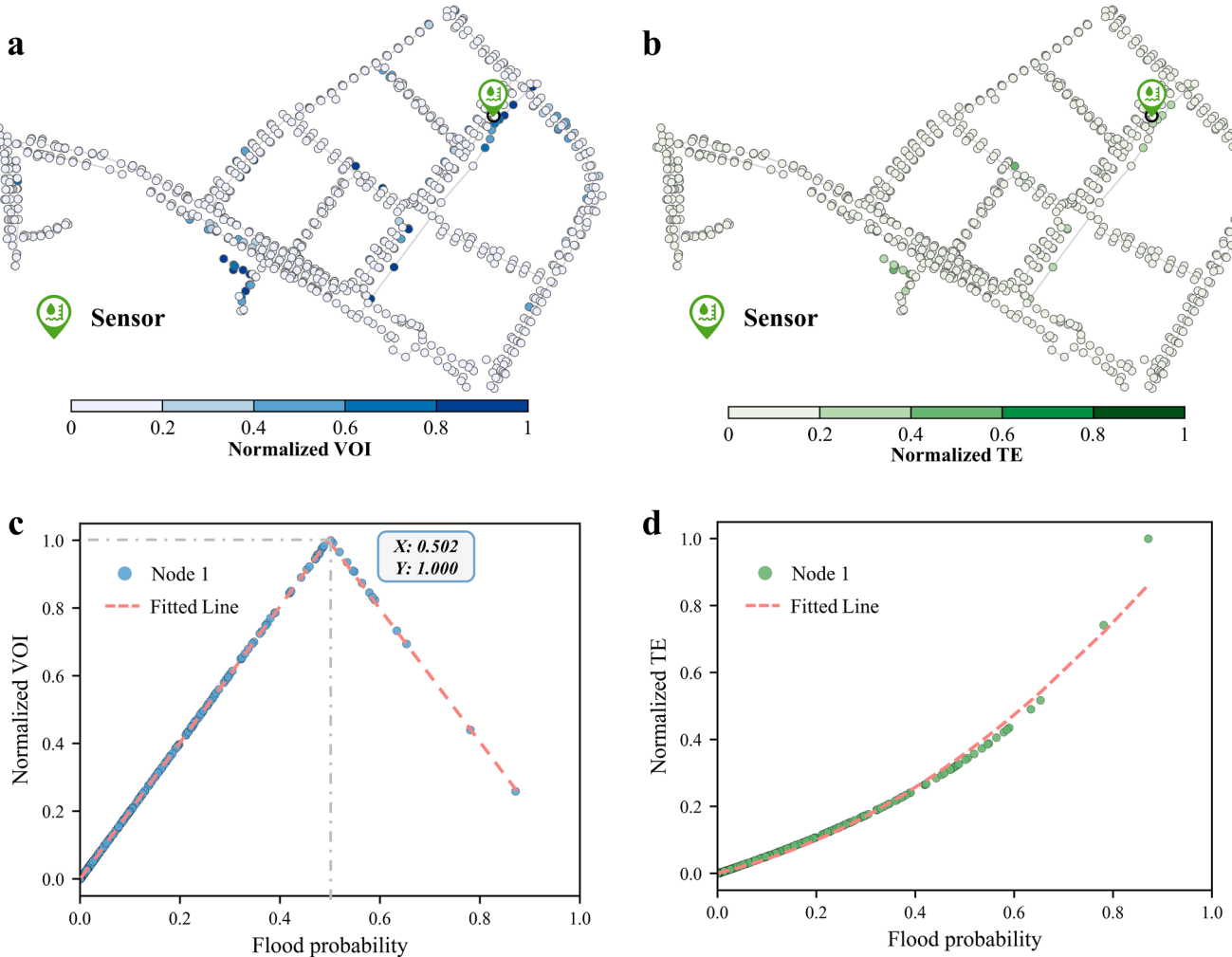


Fig. 4. (a) Normalized $VOI_1(j)$ for node 1, (b) normalized $TE(1,j)$ for node 1, (c) the relationship between flood probability and normalized $VOI_1(j)$, and (d) the relationship between flood probability and normalized $TE(1,j)$.

of 0.5 has the highest uncertainty. Specifically, the state of node j would gradually stabilize when its flooding probability closed to 0 (normal state) or 1 (flooding state), and the value of the information provided by the sensor node i would also gradually decrease due to the stability and predictability. In contrast, the state obtained from sensor node i would play a significant role in assessing the state of node j (the probability of normal state was similar to the probability of flooding state). Furthermore, the information redundancies between sensor node and other nodes were clearly demonstrated in Fig. 4 and Fig. S3-S4. Overall, the normalized TE values of each node were generally increased with the increasing of the flooding probability difference between node j and sensor node i . For example, when the flooding probability of selected sensor node i was 0.5, the normalized TE values eventually raised to 1 when the flooding probability was less than 0.5, and the normalized TE values also had a slight increasing when it was greater than 0.5. From the perspective of information redundancy, this result indicated that there was more mutual information between the sensor node i and node j , that is, the information redundancy was large when the flooding probability of them was close (with a strong correlation). Thus, the values of normalized VOI and TE can effectively represent the amount of information provided by the selected sensor node and the information redundancy between nodes.

4.2. Results of multi-objective optimization

After calculation of the VOI and TE matrices, the NSGA-II optimization model was executed to find the set of sensor layout with the maximum VOI, minimum TE, and minimum number of sensors. To provide an understanding the optimization results of different number of sensors, the obtained pareto-optimal solutions contained 1500 layouts with 2-120 nodes for the placement of sensors. Table 2 shows pareto-optimal solutions in terms of the three objectives of this study, and their corresponding values of VOI and TE. It can be seen that the VOI values had a highly positive relationship with the number of sensors, and the correlation trend was not mitigated when the number of sensors was less than 120 in this study. However, the TE values did not show similar variation characteristics, and the TE values had large differences even when VOI values were similar of the same number of sensors. For example, the Pareto points 412 and 413 with 59 nodes with VOI values equal to 58.82 and 58.89, but their TE values equal to 0.61 and 4.20,

Table 2

A randomly selected set of pareto-optimal solutions with their respective values of objective functions.

# Pareto points	Number of sensors	VOI	TE
1	2	1.99	2.0×10^{-6}
2	2	2.00	0.12
3	3	3.00	0.37
4	3	2.99	0.01
5	3	2.99	5.3×10^{-5}
6	3	2.99	0.31
⋮	⋮	⋮	⋮
412	59	58.82	0.61
413	59	58.89	4.20
414	60	59.89	12.56
415	60	59.82	0.54
⋮	⋮	⋮	⋮
732	79	78.77	7.04
733	79	78.76	0.94
734	80	79.76	0.96
735	80	79.85	16.35
⋮	⋮	⋮	⋮
1078	99	98.79	23.45
1079	99	98.71	6.61
1080	100	99.71	10.03
1081	100	99.70	6.70
⋮	⋮	⋮	⋮
1499	120	119.72	21.24
1500	120	119.64	8.66

respectively. This implies that the impact of information redundancy should be emphasized in the pareto-optimal solutions in this study.

Fig. S5-S6 depicted the relationship between the number of sensors and VOI values (objective function 1), as well as TE values (objective function 2) in the Pareto-optimal solutions. It is obvious that as the number of sensors increased, the VOI values also increased completely linearly. But the variation and distribution of TE values with the number of sensors were more complex, and the optimization model could find the node combination with a TE value close to 0 when the number of sensors was less than about 100. This result indicated that there was a threshold number of the sensors, below which a combination of sensor node with a very low level of information redundancy can be found. Therefore, it is appropriate to determine the TE value as the primary consideration for selecting optimal sensor placement in the pareto-optimal solutions, because the number of sensors installed at the study area would not be more than 100 according to the total number of nodes and economic costs in UDS.

Since one of the main purposes of implementing this methodology was to minimize the total cost of operation status perception and flooding risk assessment in UDS, the number of water level sensors played an important role (Wang et al., 2023). Thus, we constrained the selection strategy of pareto front to select sensor layouts with no more than 20. Fig. 5 shown the pareto-optimal solutions with 2-20 nodes for placement of sensors. According to the trade-off curve between number of sensors and VOI (TE) values (Fig. 5a and 5b), we selected a set of potential locations for placement of sensors contain 3 nodes to compare the three existing sensors. The VOI value of the selected potential sensor combination was 0.138 greater than the existing monitoring sensors, while the TE value was 0.0039 smaller (but still two orders of magnitude smaller) (Fig. 5c). Moreover, five placement combinations with different numbers of sensors (4, 8, 12, 16, and 20) were selected from the pareto-optimal solutions. The specific layout of these sensor networks was shown in Fig. S7.

4.3. Operation status evaluation of UDS

VOI and TE were used as the indirect metrics to find the optimal layout of sensors in UDS. However, vital direct metrics that were used for performance assessment of any designed layout are their capability of operation status perception and flooding risk assessment. Firstly, the no flooding scenario under a rainfall event was selected to evaluate the performance of designed sensor networks. We used the matrix completion approach to reduce the water level of all the nodes in UDS through sensor networks. The comparison of water level variation of existing sensor S1 and reduction data by matrix completion during the rainfall event were shown in Fig. 6a. It can be seen from Fig. 6a, despite of some deviations, the reduction results of matrix completion approach could generally restore the variation pattern of observed data of sensor S1. There were similar results of existing sensor S2 and S3 (Fig. S8), and these results proved that the matrix completion results were relatively reliable. Although the performance of matrix completion was not excellent ($NSE < 0.5$), improving the accuracy of the reduction results was not the focus of this study. Therefore, this paper did not focus further on whether the matrix completion results had high performance, but compared the reduction results of different designed sensor networks.

Then, designed optimal sensor networks with different numbers were used to calculate the water level of other nodes, and the reduction results were compared to the sensor networks with the same number of sensors which were placed randomly. As shown in Fig. 6b, while the uncertainties in the initialization of the matrix completion algorithm caused some inherent randomness, the MAPE value of designed optimal sensor networks broadly exhibited a trend of decreasing and then leveling off as the quantity of the sensor increased. However, the random selection method exhibited greater uncertainty than the optimal selection method. This result revealed that the increase in the quantity of

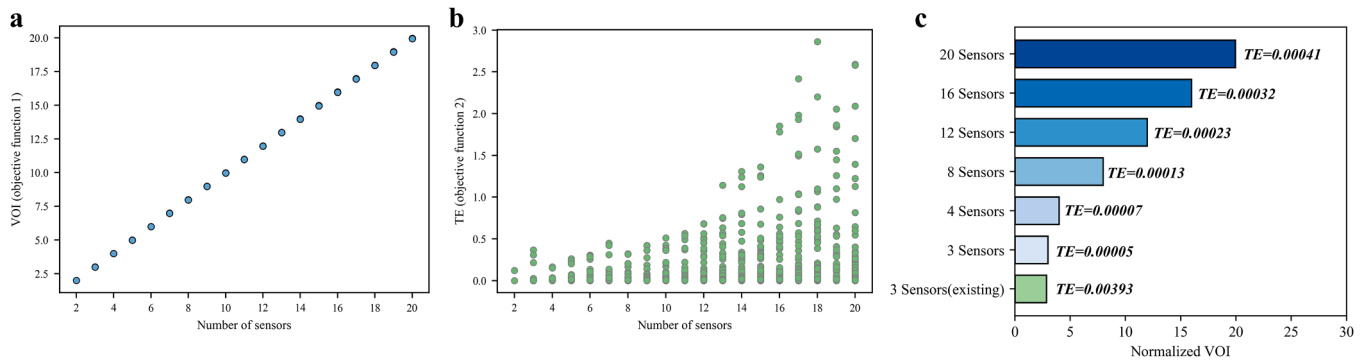


Fig. 5. The optimization results of NSGA-II model.

(a) Trade-off curve between number of sensors and VOI values, (b) Trade-off curve between number of sensors and TE values, and (c) the VOI (TE) values of existing sensor network and selected sensor network.

sensor can effectively improve the operation status perception ability of sensor network to the UDS, regardless of whether it is the random selection method or optimal selection method. This phenomenon could be attributed to the increase of the VOI value of the sensor network with the increase of sensor number. Furthermore, it can be seen that the optimal selection method significantly improved the content of effective information in sensor networks by reducing the TE value among nodes. It is worth noting that although the random selection method would lead to a significant increase in redundant information, the amount of effective information in sensor networks was still on the rise. The most significant improvement in calculation accuracy of optimal selection method occurred when the number of sensors was increased from two to three. The performance of the sensor network became temporarily stable when the number of sensors increased to four. Then the performance continued to improve until the number of sensors increased to eight, and the accuracy of matrix completion results appeared to approach the growth threshold. Quantitatively, the operation status perception error was 33 %, and reduced the error to 29 % when increasing the sensor count to eight. The amount of selected sensors was not linearly positively correlated to the performance of matrix completion but rather had an approximately logarithmic correlation. Therefore, considering the economic cost and engineering practice, setting the amount of sensors to four (lower budget) or eight (higher budget) was appropriate in this study. Municipal engineers need to design appropriate sensor network based on the characteristics of urban drainage system in the corresponding study area, which demonstrates the importance of the proposed sensor placement optimization and evaluation framework for urban flooding side-by-side.

To delve deeper into the spatial distribution of matrix completion performance, the accuracy of each node in UDS was exhibited in Fig. 6 and Fig. S9. The RMSE values of each node in UDS reduced by the designed sensor network with 4 sensors in section 4.2 were illustrated in Fig. 6c. The nodes with large errors were the river that crosses the study area, and its connected nodes. There are two possible explanations for this result. On the one hand, we simplified the river model during the modeling process and used the rectangular open conduits to approximate the small river, which would lead to significant differences in hydraulic characteristics between the simulated river and the real river. On the other hand, the river nodes were not involved in the optimal layout of sensors of this study, thus it is almost difficult to obtain effective information about the small river from the designed sensor network. Furthermore, there was no spatial dependency between the performance of each node and the location of sensors. By comparison, the same results of designed sensor network with 8 sensors were shown in Fig. 6e. It was apparent that the performance of most nodes has been improved, but the RMSE value of the river nodes increased significantly. Quantitatively, the peak of the gaussian density curve in probability density distribution is around 0.9 for designed sensor network with 4

sensors, while the RMSE values were mainly concentrated in the range of 0.8–1.6 (Fig. 6d). The peak of the gaussian density curve decreased to around 0.6 and the RMSE concentration range was 0.4–1.4 for designed sensor network with 8 sensors (Fig. 6f). These results demonstrated that the new information obtained by sensors effectively improves the performance of the sensor network when the number of sensors in the designed sensor network increased from 4 to 8. Moreover, it was reasonable that the performance of the river node has decreased due to the obtained new information was highly relevant to the nodes of UDS. Hence, it is recommended to separate the urban drainage system and the river for optimal layout of sensors in practice.

4.4. Flood risk evaluation of UDS

The ability of flooding risk assessment was also an important metric to evaluate the performance of designed sensor networks. Based on a rainfall event with serious urban flooding, we evaluated the performance of urban flooding classification under different numbers of sensors. There was a significant stage-by-stage relationship between the accuracy of under flooding and the number of sensors (Fig. 7a). Specifically, the significant improvement in classification accuracy occurred when the number of sensors increased to four or eight, and then entered a plateau period. This result indicated setting the number of sensors to four or eight is appropriate, which provided the same result as described in Section 4.3. In a real urban drainage system, most of the nodes are in a normal state without overflow, which led to the fact that the proportion of negative samples (normal state) was much larger than positive samples (flooding state) in this study. The performance of the matrix completion classification could not be measured only by accuracy, thus, the recall values under different numbers of sensor in UDS were used to evaluate the ability of sensor networks to identify positive samples (Fig. 7b). Similar to the accuracy metric, recall values also increased significantly when the number of sensors increased to four or eight. However, the most salient improvement in classification recall of optimal selection method occurred when the number of sensors was increased from two to three. This phenomenon was similar to the variation of MAPE values under different numbers of sensor. These results demonstrated that the proposed sensor placement optimization strategy successfully improved the urban flooding assessment capability of sensor networks. In addition, Fig. 7c shown the variation of recall and FPR with different numbers of sensor. The recall and FPR tended to increase alongside the growing number of selected sensors in general, and the increasing of false positive rate was within an acceptable range. As a result, the designed sensor networks could effectively reduce the risk of missed reporting of flood events, while kept the losses caused by false flood reports at a low level.

Fig. 7d illustrates the recall values spatial distribution of nodes with overflow in UDS. It is clear that the performance of partial nodes had

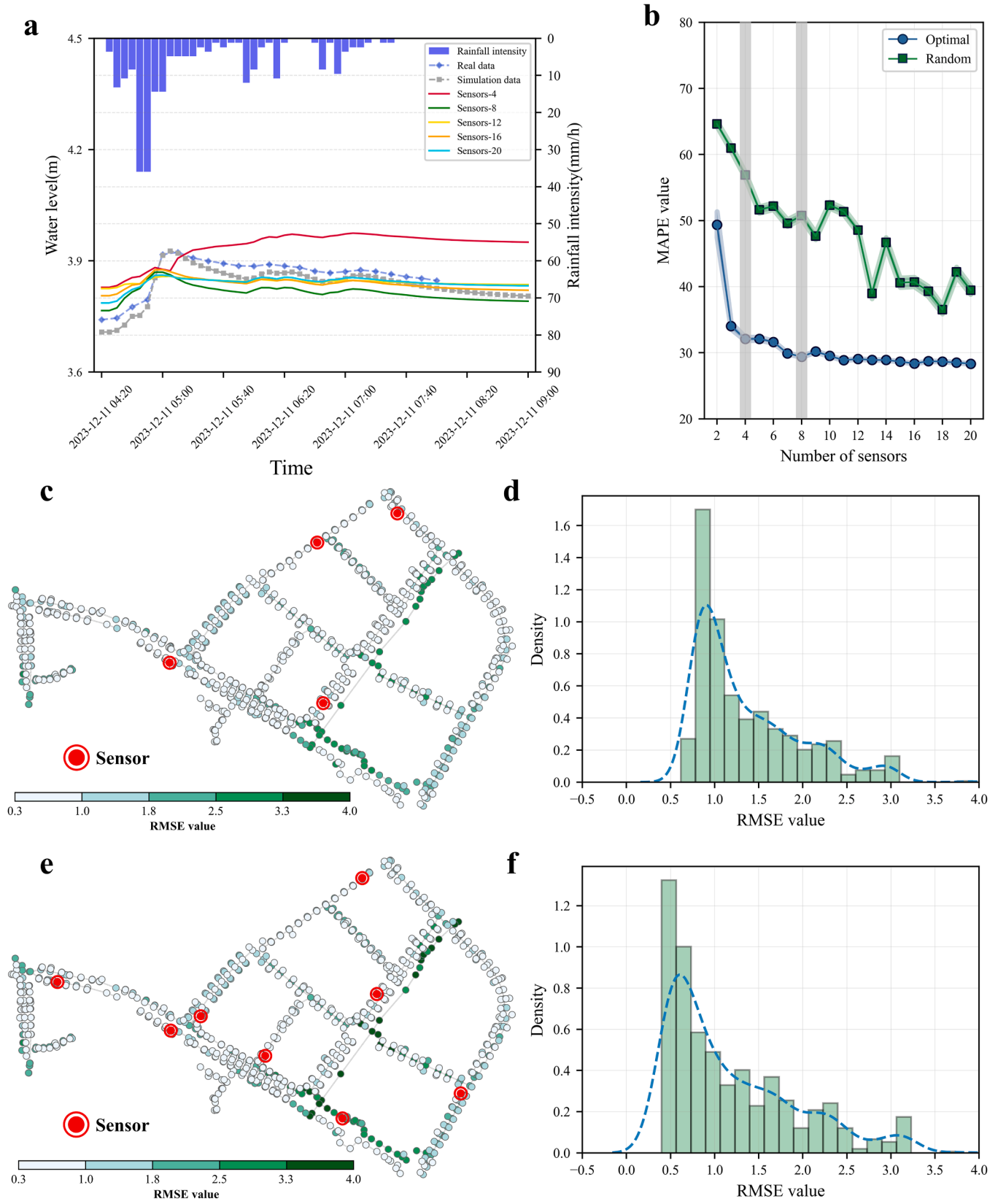


Fig. 6. The results of matrix completion under no flooding scenario. (a) The comparison between real data and reduction data (by matrix completion) of existing sensor S1, (b) The average MAPE of matrix completion results under different number of sensors, (c-d) The RMSE of each node in UDS calculated by designed sensor network with 4 sensors, (e-f) The RMSE of each node in UDS calculated by designed sensor network with 8 sensors.

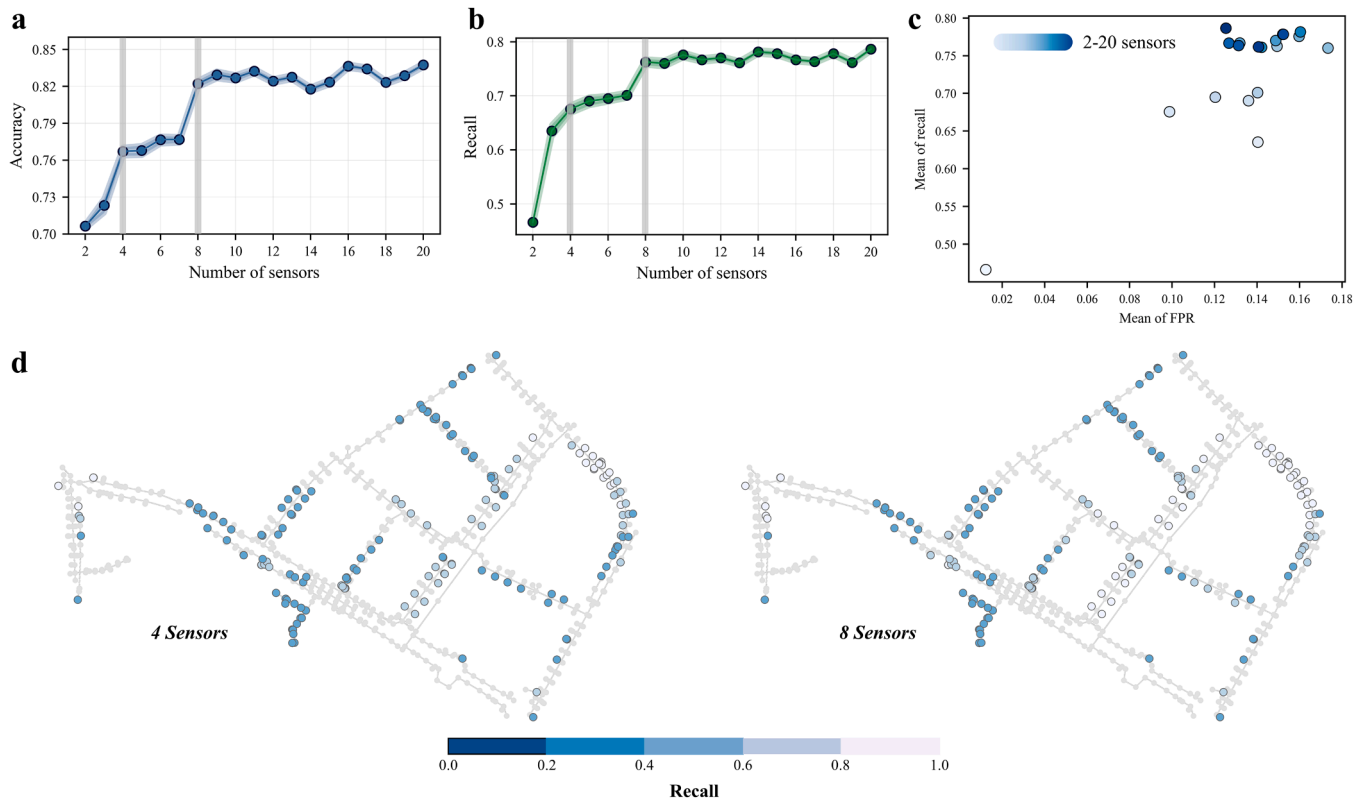


Fig. 7. The classification results of matrix completion under flooding scenario. (a) The accuracy under different numbers of sensor in UDS, (b) The recall under different numbers of sensor in UDS, (c) The relationship between FPR and recall, (d) The recall of flooding nodes in UDS under designed sensor network with 4 sensors and 8 sensors.

been significantly improved, and the positions of these nodes were relatively regular according to Fig. S10. Furthermore, as the number of sensors increased to twenty, the recall values of the remaining nodes with overflow did not obviously improve. This result may be attributed to the performance of urban flooding assessment through the matrix completion algorithm having reached a bottleneck under the limited number of sensors or the optimization model having made a balance to cope with different conditions of UDS. Results proved that the matrix completion algorithm in this study is an appropriate tool for the rapid evaluation of sensor placement optimization due to its low computation time cost.

There are some common parameters used to assess the performance of any designed layout for their capability of detecting leakage and contaminant intrusion in WDS; i.e., probability of detection, and average time to detection (Khorshidi et al., 2020; Khorshidi et al., 2018). However, previous studies have yet to provide suitable parameters or approaches for the evaluation of sensor network performance in UDS. For example, Wang et al., (2023) used information theory-based evaluation metrics to quantitatively assess the designed sensor networks. Vonach et al., (2018) found a suitable sensor network for the study area by assessing the suitability of nodes as calibration points. These methods can not rapidly quantify the capability of status perception and flooding risk assessment for the designed sensor networks. Hence, the proposed framework in this study provides a fast and effective method for sensor network deployment and evaluation of urban drainage systems.

4.5. Limitations and future work

In this work, we proposed the sensor placement optimization and evaluation framework based on information theory and matrix completion technology. The designed sensor networks through the approach in this study, significantly improved the performance of the

ability of operation status perception and flooding risk assessment. Nevertheless, the method limitations should be acknowledged, and future works are recommended in this study.

Firstly, the water level was used as the only indicator for sensor placement optimization because the aim of this study is hydraulic monitoring for urban flooding. However, there are various types of sensors in the real urban drainage systems, thus the future studies may expand the proposed framework to apply different parameters in response to the demand for routine monitoring. For example, we agree that in certain cases, considering flow data may be beneficial for flood modeling and sensor network design. Therefore, within budget constraints, we suggest installing a limited number of flow sensors at crucial points in the drainage system (e.g., discharge outlets, flood-prone areas), in conjunction with the existing water level sensors. This approach could further enhance the performance of the sensor network. In addition, the basic matrix completion technique was applied to evaluate the performance of sensor networks in this study, without any prior knowledge or the improvement of the matrix completion model. Therefore, it is recommended to explore the ability of prior knowledge such as topological relationships or technologies such as graph regularization (He et al., 2018) and deep learning (Fan and Chow, 2017) to improve the accuracy of matrix completion. These researches will help to extract the maximum information from the designed sensor networks, and improve their effectiveness in both spatial and temporal domains.

5. Conclusions

Robust sensor networks with optimized design are fundamental to ensuring the efficient and safe operation of urban drainage systems. This paper proposed a novel framework for sensor placement optimization and evaluation in UDS. Based on the case study results, the main conclusions are as follows:

- (1) We utilized the values of VOI and TE to represent the information amount provided by the selected monitoring network and the redundancy between nodes, and then the NSGA-II multi-objective optimization model was executed to identify sensor layouts with the minimum number of sensors, maximum VOI and minimum TE. The results demonstrated that the VOI and TE effectively capture the information characteristics of sensor networks. Moreover, it is found that the VOI values increased linearly with the number of selected sensors according to the pareto optimal solution results, while the TE is the main objective affecting the layouts with the same number of sensors. Consequently, we selected five sensors networks with different numbers of sensors (4, 8, 12, 16, and 20) from the pareto optimal solutions in this study.
- (2) The matrix completion algorithm successfully evaluated the sensor network's performance in operation status perception and flooding risk assessment, thereby demonstrating its significant potential in engineering applications. Results from the no flooding scenario under a rainfall event shown that the optimal selection method significantly improved performance of sensor networks compared to the random selection method in operation status perception. The results indicated that the operation status perception error was 33 %, and the flooding risk assessment accuracy reached 76 % with four sensors. Increasing the sensor count to eight reduced the error to 29 % and improved accuracy to 82 %. In addition, it is recommended to set the amount of sensors to four (lower budget) or eight (higher budget) by considering the evaluation results and economic cost. Finally, the matrix completion algorithm can also rapidly and accurately assess the flooding risk performance of the designed sensor networks during a serious urban flooding event.

In summary, the proposed sensor placement optimization and evaluation framework has proven efficient and effective in designing UDS sensor networks. The most significant feature of our approach is its ability to clearly and directly exhibit the performance of any designed sensor network. This framework provides an economical and practical method for the scientific management of urban drainage systems. We believe that this framework provides an economical and practical method for the scientific management of urban drainage systems.

CRedit authorship contribution statement

Yue Zheng: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xiaoming Jin:** Resources, Investigation, Data curation. **Jun Wei:** Supervision, Resources, Project administration. **Yongchao Zhou:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition. **Yiping Zhang:** Writing – review & editing, Supervision, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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Data availability

Data will be made available on request.

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